

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application Practical

Practical – V(Curiosity Question)

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Here is a clearer and more professional version of your algorithm:

ALGORITHM

1. Open **Visual Studio**.
2. Create a new **ASP.NET Core Web App (Model-View-Controller)** project.
3. Install the following required NuGet packages:
4. Microsoft.EntityFrameworkCore
5. Microsoft.EntityFrameworkCore.SqlServer
6. Microsoft.EntityFrameworkCore.Tools
7. Microsoft.EntityFrameworkCore.Design
8. Microsoft.VisualStudio.Web.CodeGeneration.Design
9. Microsoft.AspNetCore.Authentication.Cookies
10. Create a **Product** model with the following properties:
11. Id
12. Name
13. Category
14. Price
15. Quantity
16. Create an **AppDbContext** class that inherits from **DbContext** and add a **DbSet<Product>** property.
17. Configure the SQL Server connection string in the **appsettings.json** file.
18. Register the **AppDbContext** in **Program.cs** using the **AddDbContext()** method.
19. Configure **Cookie Authentication** and **Authorization** in **Program.cs**.
20. Create a **HomeController** with the following actions:
21. Index

22. Login (GET)
23. Login (POST)
24. Logout
25. SetTheme
26. Implement **Cookie Authentication** in the Login action using `ClaimsIdentity`, `ClaimsPrincipal`, and `SignInAsync()`.
27. Use **Cookies** to store the user's preferred theme or language.
28. Open the **Package Manager Console** and execute the following commands:

Add-Migration InitialCreate
29. Update-Database
30. In **Solution Explorer**, right-click the **Controllers** folder and select **Add** → **New Scaffolded Item**.
31. Choose **MVC Controller with views, using Entity Framework**.
32. Select the **Product** model class and **AppDbContext**, then click **Add** to automatically generate the **ProductController** and CRUD views.
33. Add the **[Authorize]** attribute to the **ProductController** so that only authenticated users can access the product management module.
34. Build and run the application.
35. Open the **Home** page and select a preferred theme to verify that it is stored using **Cookies**.
36. Navigate to the **Login** page and log in using the administrator credentials.
37. After successful login, navigate to the **Products** page.
38. Click **Create New** and enter the product details.
39. Save the record and verify that it appears in the product list.
40. Use the **Edit** option to update an existing product and save the changes.
41. Use the **Details** option to view complete information about a selected product.
42. Use the **Delete** option to remove a product from the database and confirm the deletion.
43. Verify that all CRUD (Create, Read, Update, Delete) operations are functioning correctly and that only authenticated users can access the product management module.
44. Close the application and stop debugging.

CODE:

([ProductController.cs](#)):

```
using Microsoft.AspNetCore.Authorization;
using Microsoft.AspNetCore.Mvc;
using WebApplication3.Data;
using WebApplication3.Models;
namespace WebApplication3.Controllers
{
    [Authorize]
    public class ProductController : Controller
    {
        private readonly AppDbContext db;
        public ProductController(AppDbContext context)
        {
            db = context;
        }
        public IActionResult Index()
        {
            return View(db.Products.ToList());
        }
        public IActionResult Create()
        {
            return View();
        }
        [HttpPost]
        public IActionResult Create(Product product)
        {
            db.Products.Add(product);
            db.SaveChanges();
            return RedirectToAction("Index");
        }
        public IActionResult Edit(int id)
        {
            return View(db.Products.Find(id));
        }
        [HttpPost]
        public IActionResult Edit(Product product)
        {
            db.Products.Update(product);
            db.SaveChanges();
            return RedirectToAction("Index");
        }
        public IActionResult Delete(int id)
        {
            var p = db.Products.Find(id);
            db.Products.Remove(p);
        }
    }
}
```

```

        db.SaveChanges();
        return RedirectToAction("Index");
    }
    public IActionResult Details(int id)
    {
        return View(db.Products.Find(id));
    }
}
}

```

(Program.cs):

```

using Microsoft.AspNetCore.Authentication.Cookies;
using Microsoft.EntityFrameworkCore;
using WebApplication3.Data;
var builder = WebApplication.CreateBuilder(args);
// Add services
builder.Services.AddControllersWithViews();
// Register DbContext
builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("DefaultConnection")));
// Cookie Authentication
builder.Services.AddAuthentication(CookieAuthenticationDefaults.AuthenticationScheme)
    .AddCookie(options =>
    {
        options.LoginPath = "/Home/Login";
        options.AccessDeniedPath = "/Home/Login";
    });
// Authorization
builder.Services.AddAuthorization();
var app = builder.Build();
// Configure middleware
if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
    app.UseHsts();
}
app.UseHttpsRedirection();
app.UseStaticFiles();
app.UseRouting();
app.UseAuthentication();
app.UseAuthorization();
app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}");

```

app.Run();

(appsetting.json):

```
{
  "ConnectionStrings": {
    "DefaultConnection":
      "Server=(localdb)\\MSSQLLocalDB;Database=ShoppingDB;Trusted_Connection=True;TrustServerCertificate=True;",
    "AppDbContext":
      "Server=(localdb)\\mssqllocaldb;Database=AppDbContext;Trusted_Connection=True;MultipleActiveResultSets=true"
  },
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*"
}
```

(AppDbContext.cs):

```
using Microsoft.EntityFrameworkCore;
using WebApplication3.Models;

namespace WebApplication3.Data
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options)
            : base(options)
        {
        }

        public DbSet<Product> Products { get; set; }
    }
}
```

(Product.cs):

```
using System.ComponentModel.DataAnnotations;
namespace WebApplication3.Models
{
    public class Product
    {
        public int Id { get; set; }
        [Required]
        public string Name { get; set; }
        [Required]
        public string Category { get; set; }
        public decimal Price { get; set; }
        public int Quantity { get; set; }
    }
}
```

(Index.cshtml):

```
<h2>Online Shopping Management System</h2>
<h3>Selected Theme : @ViewBag.Theme</h3>
<a asp-action="SetTheme" asp-route-theme="Light">Light Theme</a>
<br>
```

```
<br>
<a asp-action="SetTheme" asp-route-theme="Dark">Dark Theme</a>
<br>
<br>
<a asp-action="Login">Login</a>
```

(Login.cshtml):

```
<form method="post">
  Username
  <input type="text" name="username" />
  <br /><br />
  Password
  <input type="password" name="password" />
  <br /><br />
  <input type="submit" value="Login" />
</form>
<h3>@ViewBag.Message</h3>
```

WebApplication3

Home

Privacy

Online Shopping Management System

Selected Theme : Light

[Light Theme](#)

[Dark Theme](#)

[Login](#)

WebApplication3

Home

Privacy

Username

Password

Login

Edit

Product

Name

Category

Price

Quantity

Save

[Back to List](#)

Edit

Product

Name

Category

Price

Quantity

[Back to List](#)

Edit Product

Name

Category

Price

Quantity

Save

[Back to List](#)

Index

[Create New](#)

Name	Category	Price	Quantity	
Ritesh	Laptop	200000.00	2	Edit Details Delete
Himani	Mobile Phone	50000.00	3	Edit Details Delete
Saqlain	Tablets	20000.00	3	Edit Details Delete